

Chapter 0: Mathematical Preliminaries

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We set a common departure point for our Optimisation journey by recalling some of the most important mathematical prerequisites for this module. This includes basic definitions from linear algebra, topological concepts, directional derivatives, and a couple of computations that we will use throughout the term.

Vector Spaces	1
Inner Products and Norms	2
Eigenvalues and Eigenvectors	6
Basic Topological Concepts	6
Directional Derivatives and Gradients	8
Gradient and Hessian of Quadratic Functions	10

Vector Spaces

The space $\mathbb{R}^n$ is the set of n -dimensional column vectors $\mathbf{x}$ with real components endowed with the component-wise addition operator:

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix},$$



and the scalar-vector product

$$\lambda \mathbf{x} = \lambda \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \\ \vdots \\ \lambda x_n \end{pmatrix}.$$

The vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ denote the standard/canonical basis, and $\mathbf{e}$ and $\mathbf{0}$ denote all ones and all zeros column vectors, respectively.

Important subsets of $\mathbb{R}^n$.

- Nonnegative orthant:

$$\mathbb{R}_+^n = \{(x_1, x_2, \dots, x_n)^\top : x_1, x_2, \dots, x_n \geq 0\}$$

- Positive orthant:

$$\mathbb{R}_{++}^n = \{(x_1, x_2, \dots, x_n)^\top : x_1, x_2, \dots, x_n > 0\}.$$

- If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the closed line segment between $\mathbf{x}$ and $\mathbf{y}$ is given by

$$[\mathbf{x}, \mathbf{y}] = \{\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x}) : \alpha \in [0, 1]\}.$$

- The open line segment $(\mathbf{x}, \mathbf{y})$ is similarly defined as

$$(\mathbf{x}, \mathbf{y}) = \{\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x}) : \alpha \in (0, 1)\}$$

for $\mathbf{x} \neq \mathbf{y}$ and $(\mathbf{x}, \mathbf{x}) = \emptyset$.

- The unit-simplex:

$$\Delta_n = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \geq \mathbf{0}, \mathbf{e}^\top \mathbf{x} = 1\}.$$

The space $\mathbb{R}^{m \times n}$. The set of all real valued $m \times n$ matrices is denoted by $\mathbb{R}^{m \times n}$. The $n \times n$ identity matrix is denoted by $\mathbf{I}_n$. The $m \times n$ zero matrix is denoted by $\mathbf{O}_{m \times n}$.

Inner Products and Norms

Definition (Inner Product). An inner product on $\mathbb{R}^n$ is a map $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ with the following properties:

1. Symmetry: $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$ for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
2. Additivity: $\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$ for any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$.
3. Homogeneity: $\langle \lambda \mathbf{x}, \mathbf{y} \rangle = \lambda \langle \mathbf{x}, \mathbf{y} \rangle$ for any $\lambda \in \mathbb{R}$ and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
4. Positive definiteness: $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ for any $\mathbf{x} \in \mathbb{R}^n$ and $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$.



Examples:



- The usual “dot product”:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i \text{ for any } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

- The “weighted dot product”:

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{w}} = \sum_{i=1}^n w_i x_i y_i$$

where $\mathbf{w} \in \mathbb{R}_{++}^n$.

Definition (Vector Norms). A norm $\|\cdot\|$ on $\mathbb{R}^n$ is a function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfying:

1. *Nonnegativity*: $\|\mathbf{x}\| \geq 0$ for any $\mathbf{x} \in \mathbb{R}^n$ and $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$.
2. *Positive homogeneity*: $\|\lambda \mathbf{x}\| = |\lambda| \|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$.
3. *Triangle inequality*: $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

A natural way to generate a norm on $\mathbb{R}^n$ is to take any inner product $\langle \cdot, \cdot \rangle$ defined on $\mathbb{R}^n$, and define the associated norm

$$\|\mathbf{x}\| \equiv \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}, \text{ for all } \mathbf{x} \in \mathbb{R}^n.$$

The norm associated with the dot-product is the so-called Euclidean norm or ℓ_2 -norm:

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2} \text{ for all } \mathbf{x} \in \mathbb{R}^n.$$

The ℓ_p -norm ($p \geq 1$) is defined by

$$\|\mathbf{x}\|_p \equiv \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

The ℓ_∞ -norm is

$$\|\mathbf{x}\|_\infty \equiv \max_{i=1,2,\dots,n} |x_i|.$$

It can be shown that

$$\|\mathbf{x}\|_\infty = \lim_{p \rightarrow \infty} \|\mathbf{x}\|_p.$$

Why $\ell_{1/2}$ is not a norm?



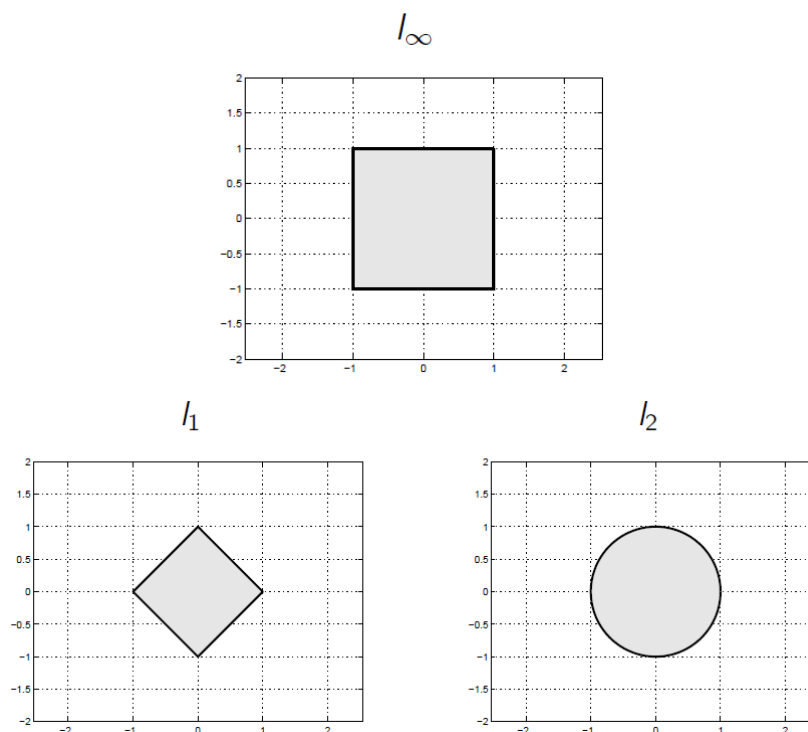


Figure 1: Different unit balls $\|\mathbf{x}\|_p \leq 1$ in $\mathbb{R}^2$.

Theorem (Cauchy-Schwartz Inequality). For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$

$$|\mathbf{x}^\top \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$$

Proof. For any $\lambda \in \mathbb{R}$:

$$\|\mathbf{x} + \lambda \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\lambda \langle \mathbf{x}, \mathbf{y} \rangle + \lambda^2 \|\mathbf{y}\|^2,$$

leading to (why?) 

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \|\mathbf{x}\|^2 \|\mathbf{y}\|^2,$$

establishing the desired result. □

Definition (Matrix Norms). A norm $\|\cdot\|$ on $\mathbb{R}^{m \times n}$ is a function $\|\cdot\| : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ satisfying

1. Nonnegativity: $\|\mathbf{A}\| \geq 0$ for any $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\|\mathbf{A}\| = 0$ if and only if $\mathbf{A} = \mathbf{0}$.
2. Positive homogeneity: $\|\lambda \mathbf{A}\| = |\lambda| \|\mathbf{A}\|$ for any $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\lambda \in \mathbb{R}$.
3. Triangle inequality: $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$ for any $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$.



Induced Norms. Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ and two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively, the induced matrix norm $\|\mathbf{A}\|_{a,b}$ (called (a,b) -norm) is defined by

$$\|\mathbf{A}\|_{a,b} = \max_{\mathbf{x}} \{\|\mathbf{A}\mathbf{x}\|_b : \|\mathbf{x}\|_a \leq 1\},$$

from where it follows that

$$\|\mathbf{A}\mathbf{x}\|_b \leq \|\mathbf{A}\|_{a,b} \|\mathbf{x}\|_a.$$

An induced norm is a norm (satisfies nonnegativity, positive homogeneity and triangle inequality). We refer to the matrix-norm $\|\cdot\|_{a,b}$ as the (a,b) -norm. When $a = b$, we will simply refer to it as an a -norm.

Relevant cases:

- **The spectral norm.** If $\|\cdot\|_a = \|\cdot\|_b = \|\cdot\|_2$, the induced $(2,2)$ -norm of a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the maximum singular value of $\mathbf{A}$

$$\|\mathbf{A}\|_2 = \|\mathbf{A}\|_{2,2} = \sqrt{\lambda_{\max}(\mathbf{A}^\top \mathbf{A})} \equiv \sigma_{\max}(\mathbf{A}).$$

This norm is called the spectral norm.

- **The ℓ_1 -norm.** When $\|\cdot\|_a = \|\cdot\|_b = \|\cdot\|_1$, the induced $(1,1)$ -matrix norm of a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ is given by

$$\|\mathbf{A}\|_1 = \max_{j=1,2,\dots,n} \sum_{i=1}^m |A_{i,j}|.$$

- **The ℓ_∞ -norm.** When $\|\cdot\|_a = \|\cdot\|_b = \|\cdot\|_\infty$, the induced (∞, ∞) -matrix norm of matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ is given by

$$\|\mathbf{A}\|_\infty = \max_{i=1,2,\dots,m} \sum_{j=1}^n |A_{i,j}|$$

- **The Frobenius norm.**

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2}, \quad \mathbf{A} \in \mathbb{R}^{m \times n}$$

The Frobenius norm is not an induced norm. Why is it a norm?



Eigenvalues and Eigenvectors

Definition (Eigenvalues and eigenvectors). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. Then a nonzero vector $\mathbf{v} \in \mathbb{R}^n$ is called an eigenvector of $\mathbf{A}$ if there exists a $\lambda \in \mathbb{C}$ for which

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

The scalar λ is the eigenvalue corresponding to the eigenvector $\mathbf{v}$.

In general, real-valued matrices can have complex eigenvalues, but when the matrix is symmetric the eigenvalues are necessarily real. The eigenvalues of a symmetric $n \times n$ matrix $\mathbf{A}$ are denoted by

$$\lambda_1(\mathbf{A}) \geq \lambda_2(\mathbf{A}) \geq \dots \geq \lambda_n(\mathbf{A}).$$

The maximum eigenvalue is also denoted by $\lambda_{\max}(\mathbf{A}) (= \lambda_1(\mathbf{A}))$ and the minimum eigenvalue is also denoted by $\lambda_{\min}(\mathbf{A}) (= \lambda_n(\mathbf{A}))$.

Theorem (Spectral Factorization Theorem). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be an $n \times n$ symmetric matrix. Then there exists an orthogonal matrix $\mathbf{U} \in \mathbb{R}^{n \times n}$ ($\mathbf{U}^T \mathbf{U} = \mathbf{U} \mathbf{U}^T = \mathbf{I}_n$) and a diagonal matrix $\mathbf{D} = \text{diag}(d_1, d_2, \dots, d_n)$ for which

$$\mathbf{U}^T \mathbf{A} \mathbf{U} = \mathbf{D}.$$

The columns of the matrix $\mathbf{U}$ constitute an orthogonal basis comprising eigenvectors of $\mathbf{A}$ and the diagonal elements of $\mathbf{D}$ are the corresponding eigenvalues. A direct result is that $\text{Tr}(\mathbf{A}) = \sum_{i=1}^n \lambda_i(\mathbf{A})$ and $\det(\mathbf{A}) = \prod_{i=1}^n \lambda_i(\mathbf{A})$.

Another important consequence of the spectral decomposition theorem is the bounding of the so-called Rayleigh quotient. For a symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, the Rayleigh quotient is defined by

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\|\mathbf{x}\|^2} \text{ for any } \mathbf{x} \neq 0$$

We use the spectral decomposition theorem to show that if $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, then

$$\lambda_{\min}(\mathbf{A}) \leq R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_{\max}(\mathbf{A}) \text{ for any } \mathbf{x} \neq 0.$$

Basic Topological Concepts

The open ball with center $c \in \mathbb{R}^n$ and radius r :

$$B(c, r) = \{\mathbf{x} : \|\mathbf{x} - c\| < r\}.$$

The closed ball with center c and radius r :

$$B[c, r] = \{\mathbf{x} : \|\mathbf{x} - c\| \leq r\}.$$



Definition (Interior Point). Given a set $U \subseteq \mathbb{R}^n$, a point $\mathbf{c} \in U$ is called an interior point of U if there exists $r > 0$ for which $B(\mathbf{c}, r) \subseteq U$. The set of all interior points of a given set U is called the interior of the set and is denoted by $\text{int}(U)$:

$$\text{int}(U) = \{\mathbf{x} \in U : B(\mathbf{x}, r) \subseteq U \text{ for some } r > 0\}.$$

Examples:



$$\begin{aligned}\text{int}(\mathbb{R}_+^n) &= \mathbb{R}_{++}^n \\ \text{int}(B[\mathbf{c}, r]) &= B(\mathbf{c}, r) \quad (\mathbf{c} \in \mathbb{R}^n, r \in \mathbb{R}_{++}) \\ \text{int}([x, y]) &=?\end{aligned}$$

An open set is a set that contains only interior points. Meaning that

$$U = \text{int}(U).$$

Examples of open sets are open balls (hence the name...) and the positive orthant $\mathbb{R}_{++}^n$. The union of any number of open sets is an open set and the intersection of a finite number of open sets is open.

Closed Sets. A set $U \subseteq \mathbb{R}^n$ is closed if it contains all the limits of convergent sequences of vectors in U , that is, if $\{\mathbf{x}_i\}_{i=1}^\infty \subseteq U$ satisfies $\mathbf{x}_i \rightarrow \mathbf{x}^*$ as $i \rightarrow \infty$, then $\mathbf{x}^* \in U$. A known result states that U is closed iff its complement U^c is open. Examples of closed sets are the closed ball $B[\mathbf{c}, r]$, closed line segments, the nonnegative orthant $\mathbb{R}_+^n$ and the unit simplex Δ_n . What about $\mathbb{R}^n$? $\emptyset$?

Boundary Points. Given a set $U \subseteq \mathbb{R}^n$, a boundary point of U is a vector $\mathbf{x} \in \mathbb{R}^n$ satisfying the following: any neighbourhood of $\mathbf{x}$ contains at least one point in U and at least one point in its complement U^c . The set of all boundary points of a set U is denoted by $\text{bd}(U)$.

Examples:



$$\begin{aligned}(\mathbf{c} \in \mathbb{R}^n, r \in \mathbb{R}_{++}), \text{bd}(B(\mathbf{c}, r)) &= \\ (\mathbf{c} \in \mathbb{R}^n, r \in \mathbb{R}_{++}), \text{bd}(B[\mathbf{c}, r]) &= \\ \text{bd}(\mathbb{R}_{++}^n) &= \\ \text{bd}(\mathbb{R}_+^n) &= \\ \text{bd}(\mathbb{R}^n) &= \\ \text{bd}(\Delta_n) &= \end{aligned}$$



Definition (Closure). The closure of a set $U \subseteq \mathbb{R}^n$ is denoted by $\text{cl}(U)$ and is defined to be the smallest closed set containing U :

$$\text{cl}(U) = \bigcap \{T : U \subseteq T, T \text{ is closed} \}.$$

Another equivalent definition of $\text{cl}(U)$ is:

$$\text{cl}(U) = U \cup \text{bd}(U)$$

Examples:



$$\begin{aligned} \text{cl}(\mathbb{R}_{++}^n) &= \\ (\mathbf{x} \neq \mathbf{y}), \text{cl}((\mathbf{x}, \mathbf{y})) &= \end{aligned}$$

Boundedness and Compactness. A set $U \subseteq \mathbb{R}^n$ is called bounded if there exists $M > 0$ for which $U \subseteq B(0, M)$. A set $U \subseteq \mathbb{R}^n$ is called compact if it is closed and bounded. Examples of compact sets are: closed balls, unit simplex, and closed line segments.

Directional Derivatives and Gradients

Definition (Directional Derivative). Let f be a function defined on a set $S \subseteq \mathbb{R}^n$. Let $\mathbf{x} \in \text{int}(S)$ and let $\mathbf{d} \in \mathbb{R}^n$. If the limit

$$\lim_{t \rightarrow 0^+} \frac{f(\mathbf{x} + t\mathbf{d}) - f(\mathbf{x})}{t}$$

exists, then it is called the directional derivative of f at $\mathbf{x}$ along the direction $\mathbf{d}$ and is denoted by $f'(\mathbf{x}; \mathbf{d})$. For any $i = 1, 2, \dots, n$, if the limit

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{e}_i) - f(\mathbf{x})}{t}$$

exists, then its value is called the i -th partial derivative and is denoted by $\frac{\partial f}{\partial x_i}(\mathbf{x})$. If all the partial derivatives of a function f exist at a point $\mathbf{x} \in \mathbb{R}^n$, then the gradient of f at $\mathbf{x}$ is

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \frac{\partial f}{\partial x_2}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}) \end{pmatrix}.$$

Definition (Continuous Differentiability). A function f defined on an open set $U \subseteq \mathbb{R}^n$ is called continuously differentiable over U if all the partial derivatives exist and are continuous on U . In that case,

$$f'(\mathbf{x}; \mathbf{d}) = \nabla f(\mathbf{x})^\top \mathbf{d}, \quad \mathbf{x} \in U, \mathbf{d} \in \mathbb{R}^n.$$



Proposition. Let $f : U \rightarrow \mathbb{R}$ be defined on an open set $U \subseteq \mathbb{R}^n$. Suppose that f is continuously differentiable over U . Then

$$\lim_{\mathbf{d} \rightarrow 0} \frac{f(\mathbf{x} + \mathbf{d}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{d}}{\|\mathbf{d}\|} = 0, \quad \text{for all } \mathbf{x} \in U.$$

Another way to write the above result is as follows:

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + o(\|\mathbf{y} - \mathbf{x}\|)$$

where $o(\cdot) : \mathbb{R}_+^n \rightarrow \mathbb{R}$ is a one-dimensional function satisfying $\frac{o(t)}{t} \rightarrow 0$ as $t \rightarrow 0^+$.

Twice Differentiability and the Hessian

The partial derivatives $\frac{\partial f}{\partial x_i}$ are themselves real-valued functions that can be partially differentiated. The (i, j) -partial derivatives of f at $\mathbf{x} \in U$ (if exists) is defined by

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) = \frac{\partial \left(\frac{\partial f}{\partial x_j} \right)}{\partial x_i}(\mathbf{x}).$$

A function f defined on an open set $U \subseteq \mathbb{R}^n$ is called twice continuously differentiable over U if all the second order partial derivatives exist and are continuous over U . In that case, for any $i \neq j$ and any $\mathbf{x} \in U$:

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x})$$

Definition (The Hessian). The Hessian of f at a point $\mathbf{x} \in U$ is the $n \times n$ matrix:

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & & \vdots \\ \vdots & \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

For twice continuously differentiable functions, the Hessian is a symmetric matrix.

Theorem (Linear Approximation Theorem). Let $f : U \rightarrow \mathbb{R}$ be defined on an open set $U \subseteq \mathbb{R}^n$. Suppose that f is twice continuously differentiable over U . Let $\mathbf{x} \in U$ and $r > 0$ satisfy $B(\mathbf{x}, r) \subseteq U$. Then for any $\mathbf{y} \in B(\mathbf{x}, r)$, there exists $\xi \in [\mathbf{x}, \mathbf{y}]$ such that:

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{1}{2} (\mathbf{y} - \mathbf{x})^\top \nabla^2 f(\xi) (\mathbf{y} - \mathbf{x}).$$



Theorem (Quadratic Approximation Theorem). Let $f : U \rightarrow \mathbb{R}$ be defined on an open set $U \subseteq \mathbb{R}^n$. Suppose that f is twice continuously differentiable over U . Let $\mathbf{x} \in U$ and $r > 0$ satisfy $B(\mathbf{x}, r) \subseteq U$. Then for any $\mathbf{y} \in B(\mathbf{x}, r)$:

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{1}{2} (\mathbf{y} - \mathbf{x})^\top \nabla^2 f(\mathbf{x}) (\mathbf{y} - \mathbf{x}) + o(\|\mathbf{y} - \mathbf{x}\|^2) .$$

The little “ o ” notation above can be interpreted as $o(\|\mathbf{x} - \mathbf{y}\|^2)$ goes faster to zero than $\|\mathbf{x} - \mathbf{y}\|^2$.

Gradient and Hessian of Quadratic Functions

Gradient of Linear Function

Consider a linear function of the form

$$f(\mathbf{w}) = \mathbf{a}^\top \mathbf{w}$$

where $\mathbf{a}$ and $\mathbf{w} \in \mathbb{R}^n$. We can derive the gradient in matrix notation as follows:

1. Convert to summation notation:

$$f(\mathbf{w}) = \sum_{j=1}^n a_j w_j .$$

2. Take the partial derivative with respect to a generic element k :

$$\frac{\partial}{\partial w_k} \left[\sum_{j=1}^n a_j w_j \right] = a_k .$$

3. Assemble the partial derivatives into a vector:

$$\nabla f(\mathbf{w}) = \begin{bmatrix} \frac{\partial}{\partial w_1} \\ \frac{\partial}{\partial w_2} \\ \vdots \\ \frac{\partial}{\partial w_n} \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \mathbf{a}$$

So our final result is that

$$\nabla f(\mathbf{w}) = \mathbf{a} .$$

This generalizes the scalar case where $\frac{d}{dw} [\alpha w] = \alpha$. We can also consider general linear functions of the form

$$f(\mathbf{w}) = \mathbf{a}^\top \mathbf{w} + \beta$$

for a scalar β , but in this case we still have $\nabla f(\mathbf{w}) = \mathbf{a}$ since β does not depend on $\mathbf{w}$.



Gradient of Quadratic Function

Consider a quadratic function of the form

$$f(\mathbf{w}) = \mathbf{w}^\top A \mathbf{w}$$

where $\mathbf{w} \in \mathbb{R}^n$ and A is a matrix in $\mathbb{R}^{n \times n}$. We can derive the gradient in matrix notation as follows:

1. Convert to summation notation:

$$f(\mathbf{w}) = \mathbf{w}^\top \begin{bmatrix} \sum_{j=1}^n a_{1j} w_j \\ \sum_{j=1}^n a_{2j} w_j \\ \vdots \\ \sum_{j=1}^n a_{nj} w_j \end{bmatrix} = \sum_{i=1}^n \sum_{j=1}^n w_i a_{ij} w_j$$

where a_{ij} is the element in row i and column j of A . To help with computing the partial derivatives, it is useful to re-write it in the form

$$f(\mathbf{w}) = \sum_{i=1}^n \sum_{j=1}^n w_i a_{ij} w_j = \sum_{i=1}^n \left(a_{ii} w_i^2 + \sum_{j \neq i} w_i a_{ij} w_j \right)$$

2. Take the partial derivative with respect to a generic element k :

$$\frac{\partial}{\partial w_k} \left[\sum_{i=1}^n \left(a_{ii} w_i^2 + \sum_{j \neq i} w_i a_{ij} w_j \right) \right] = 2a_{kk} w_k + \sum_{j \neq k} w_j a_{jk} + \sum_{j \neq k} a_{kj} w_j$$

The first term comes from the a_{kk} term that is quadratic in w_k , while the two sums come from the terms that are linear in w_k . We can move one $a_{kk} w_k$ into each of the sums to simplify this to

$$\frac{\partial}{\partial w_k} \left[\sum_{i=1}^n \left(a_{ii} w_i^2 + \sum_{j \neq i} w_i a_{ij} w_j \right) \right] = \sum_{j=1}^n w_j a_{jk} + \sum_{j=1}^n a_{kj} w_j$$

3. Assemble the partial derivatives into a vector:

$$\nabla f(\mathbf{w}) = \begin{bmatrix} \frac{\partial}{\partial w_1} \\ \frac{\partial}{\partial w_2} \\ \vdots \\ \frac{\partial}{\partial w_n} \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^n w_j a_{j1} + \sum_{j=1}^n a_{1j} w_j \\ \sum_{j=1}^n w_j a_{j2} + \sum_{j=1}^n a_{2j} w_j \\ \vdots \\ \sum_{j=1}^n w_j a_{jn} + \sum_{j=1}^n a_{nj} w_j \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^n w_j a_{j1} \\ \sum_{j=1}^n w_j a_{j2} \\ \vdots \\ \sum_{j=1}^n w_j a_{jn} \end{bmatrix} + \begin{bmatrix} \sum_{j=1}^n a_{1j} w_j \\ \sum_{j=1}^n a_{2j} w_j \\ \vdots \\ \sum_{j=1}^n a_{nj} w_j \end{bmatrix}$$

4. Convert to matrix notation:

$$\nabla f(\mathbf{w}) = \begin{bmatrix} \sum_{j=1}^n w_j a_{j1} \\ \sum_{j=1}^n w_j a_{j2} \\ \vdots \\ \sum_{j=1}^n w_j a_{jn} \end{bmatrix} + \begin{bmatrix} \sum_{j=1}^n a_{1j} w_j \\ \sum_{j=1}^n a_{2j} w_j \\ \vdots \\ \sum_{j=1}^n a_{nj} w_j \end{bmatrix} = A^\top \mathbf{w} + A \mathbf{w} = (A^\top + A) \mathbf{w}$$



So our final result is that

$$\nabla f(\mathbf{w}) = (A^\top + A) \mathbf{w}$$

Note that if A is symmetric ($A^\top = A$) then we have $(A^\top + A) = (A + A) = 2A$ so we have

$$\nabla f(\mathbf{w}) = 2A\mathbf{w}$$

This generalizes the scalar case where $\frac{d}{dw} [\alpha w^2] = 2\alpha w$. We can also consider general quadratic functions of the form

$$f(\mathbf{w}) = \frac{1}{2} \mathbf{w}^\top A \mathbf{w} + \mathbf{b}^\top \mathbf{w} + \gamma$$

Using the above results we have

$$\nabla f(\mathbf{w}) = \frac{1}{2} (A^\top + A) \mathbf{w} + \mathbf{b}$$

and if A is symmetric then

$$\nabla f(\mathbf{w}) = A\mathbf{w} + \mathbf{b}$$

Hessian of a Quadratic Form

For a quadratic function of the form,

$$f(\mathbf{w}) = \mathbf{w}^\top A \mathbf{w}$$

we have shown that the partial derivatives are given by linear functions,

$$\frac{\partial f}{\partial w_k} = \sum_{j=1}^n w_j a_{jk} + \sum_{j=1}^n a_{kj} w_j.$$

The second partial derivatives are thus constant functions of the form

$$\frac{\partial^2 f}{\partial w_k \partial w_{k'}} = a_{k'k} + a_{kk'}$$

which means that the Hessian matrix has a simple form

$$\nabla^2 f(\mathbf{w}) = \begin{bmatrix} \frac{\partial}{\partial w_1} \frac{\partial}{\partial w_1} f(\mathbf{w}) & \frac{\partial}{\partial w_1} \frac{\partial}{\partial w_2} f(\mathbf{w}) & \cdots & \frac{\partial}{\partial w_1} \frac{\partial}{\partial w_n} f(\mathbf{w}) \\ \frac{\partial}{\partial w_2} \frac{\partial}{\partial w_1} f(\mathbf{w}) & \frac{\partial}{\partial w_2} \frac{\partial}{\partial w_2} f(\mathbf{w}) & \cdots & \frac{\partial}{\partial w_2} \frac{\partial}{\partial w_n} f(\mathbf{w}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial w_n} \frac{\partial}{\partial w_1} f(\mathbf{w}) & \frac{\partial}{\partial w_n} \frac{\partial}{\partial w_2} f(\mathbf{w}) & \cdots & \frac{\partial}{\partial w_n} \frac{\partial}{\partial w_n} f(\mathbf{w}) \end{bmatrix} = \begin{bmatrix} a_{11} + a_{11} & a_{21} + a_{12} & \cdots & a_{n1} + a_{1n} \\ a_{12} + a_{21} & a_{22} + a_{22} & \cdots & a_{n2} + a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} + a_{n1} & a_{2n} + a_{n2} & \cdots & a_{nn} + a_{nn} \end{bmatrix}$$

This gives a result of

$$\nabla^2 f(\mathbf{w}) = A + A^\top,$$

and if A is symmetric this simplifies to

$$\nabla^2 f(\mathbf{w}) = 2A.$$

